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SHORT REPORT

Otolith Sr:Ca ratios of freshwater goby *Rhinogobius* sp. TO indicating absence of sea migrating traits

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Abstract The strontium and calcium concentration ratios in the otoliths of *Rhinogobius* sp. TO collected from the river flowing from a pond of Aichi Expo Park, a waterway from the Yahagi and Kushida Rivers and the Yahagifuru River, Japan, were studied to confirm the presence or absence of sea migrating traits during the larval period. The ratios of all samples showed consistently low ratios from the core to the edge, suggesting that the *Rhinogobius* sp. TO has a freshwater resident lifestyle. Therefore, conservation of river ecosystems is essential to protect the natural populations of this goby.

Keywords Freshwater goby · *Rhinogobius* sp. TO · Migration · Otolith · Sr:Ca ratios

Introduction

The *Rhinogobius* species of freshwater goby is distributed widely in the Japanese Archipelago, and more than 13 species have been reported (Suzuki and Shibukawa 2004). These species have been grouped into four life history-based categories, including fluvial, amphidromous, landlocked, and lentic types (Mizuno 1961; Akihito et al. 2002; Takahashi and Okazaki 2002; Tsunagawa et al. 2010). Fluvial species like *Rhinogobius flumineus* spend their whole life in a freshwater environment (Mizuno 1961; Akihito et al. 2002). The lentic species like *Rhinogobius* sp. OR morphotype "Shimahire" and *Rhinogobius* sp. BW inhabiting a freshwater environment do not make the seaward migration and lake-river migration, respectively (Tsunagawa et al. 2010; Takahashi and Okazaki 2002). Amphidromous species like *Rhinogobius giurinus* are swept downstream to the sea after hatching in a freshwater area and return to the river after a pelagic life (Mizuno 1961; Akihito et al. 2002). In addition, some fishes of amphidromous *Rhinogobius* species have a landlocked life history in lakes, freshwater marshes, and reservoirs above dams (Akihito et al. 2002; Sakai et al. 2004; Tsunagawa and Arai 2008).

The *Rhinogobius* sp. TO (sensu Suzuki and Sakamoto 2005; Japanese name: Toukai-yoshinobori) is a recently recorded species whose distribution is limited to the periphery of the Noubi and Okazaki plains in Aichi, Gifu, and Mie prefectures (Suzuki and Sakamoto 2005; Ministry of Land, Infrastructure and Transport 2006). It appears to have a lentic lifestyle, inhabiting lentic water areas such as ponds, waterways, and slower current areas along river banks (Suzuki and Sakamoto 2005). Since habitat disappearance has been caused by artificial river improvement and landfill of ponds (Suzuki and Sakamoto 2005), practical measures must be taken to protect the natural populations of this goby. Ecological information about this goby is vital to implementing effective protective measures, but little is known about their migration history, specifically whether this fish runs down to the sea during the larval period.

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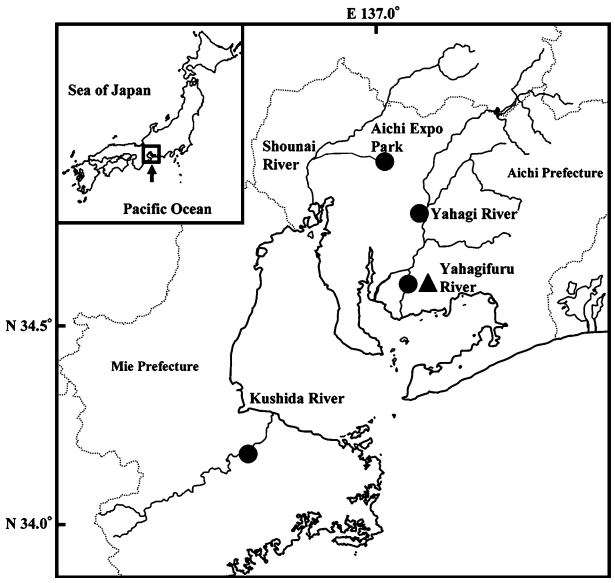


Fig. 1 Sampling sites of two *Rhinogobius* species in Aichi and Mie prefectures, Japan: *Rhinogobius* sp. TO (*circle*) and *Rhinogobius giurinus* (*triangle*)

The migratory history of diadromous fishes can be reconstructed through otolith microchemistry, particularly via the utilization of strontium to calcium concentration ratios (Campana 1999; Arai 2002). Fresh water contains low levels of Sr, while sea water contains high levels. Although the values of Sr:Ca ratios in otoliths are highly variable among species, ambient salinity is positively related to the Sr:Ca ratios of almost all fish species studied (Secor and Rooker 2000). Recently, Tsunagawa and Arai (2008, 2009) found that the otolith Sr:Ca ratios were positively related with ambient water salinity in *Rhinogobius* species. Therefore, this technique can be applied to reveal the migratory patterns of *Rhinogobius* sp. TO.

The objective of the present study was to examine whether the *Rhinogobius* sp. TO migrate to the sea during the larval period by otolith microchemical analysis. The results provide a basis of the life history in this species as well as information that should prove useful for attempts to protect them.

Materials and methods

Fish. The *Rhinogobius* fishes used in this study were collected from four locations in Aichi and Mie prefectures, Japan, from 18 October 2008 to 9 May 2009 using dip nets (Fig. 1; Table 1). *Rhinogobius* sp. TO were collected from the river flowing from a pond of Aichi Expo Park (Shounai River system) in a waterway from the Yahagi River and the Kushida River, and from the mainstream of the Yahagifuru River. *Rhinogobius giurinus* were also collected from a sympatric habitat with *Rhinogobius* sp. TO in the Yahagifuru River. All sampling sites were freshwater environments with no influence from the rising tide. After collection, all specimens were fixed immediately with 99% ethanol. A total of 31 specimens were used for the otolith microchemical analysis (Table 1).

Classification. *Rhinogobius* sp. TO was distinguished from other congeneric species according to morphological differences. Details of the classification method have been described in Suzuki and Sakamoto (2005). Furthermore, identification of *Rhinogobius giurinus* followed Akihito et al. (2002).

Otolith preparation and otolith X-ray microprobe analysis. Sagittal otoliths were ground and polished for otolith microchemical analysis after we measured the standard length of each fish. The otoliths from all specimens were used for life-history transect analysis of Sr and Ca concentrations, which were measured along a line down the longest axis of each otolith from the core to the edge using a wavelength dispersive X-ray electron microprobe (JEOL JXA-8900R). Details of both otolith preparation and microchemical analysis have been described by Tsunagawa and Arai (2008, 2009) and Tsunagawa et al. (2010).

Table 1 *Rhinogobius* sp. TO and *Rhinogobius giurinus* specimens used for otolith microchemistry analysis

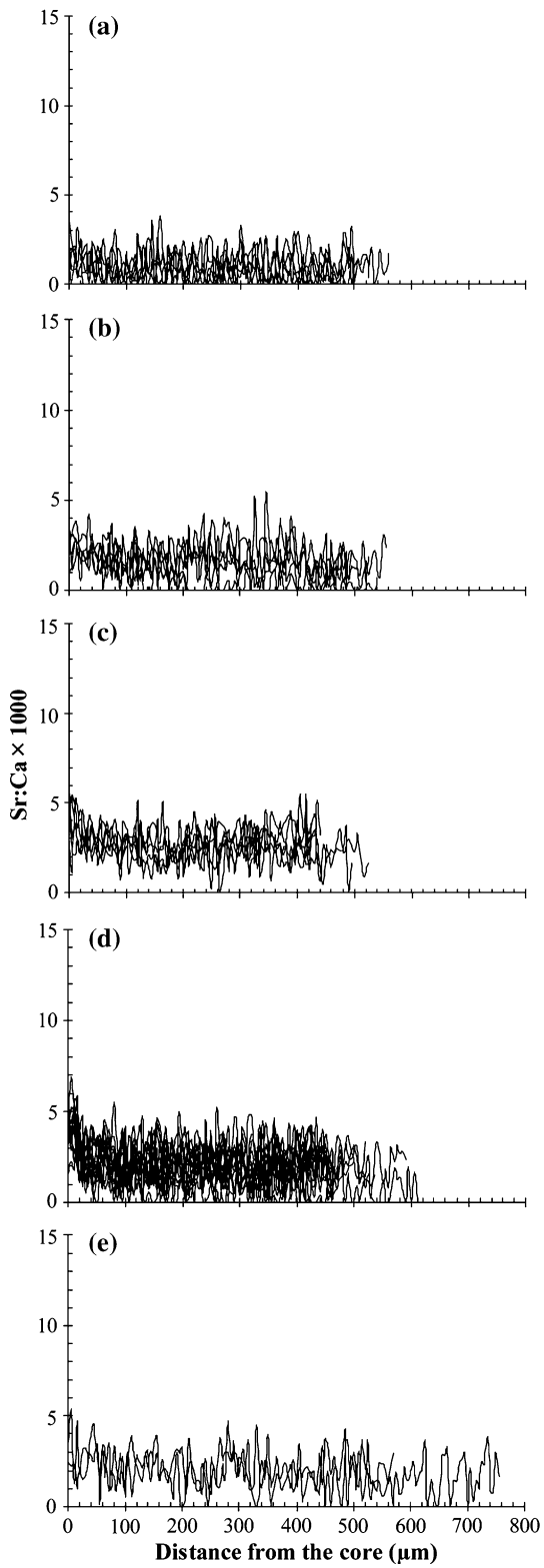
Species	River system	Microhabitat	Locality	Sampling date	Number of fish examined	Standard length (mm)
Rhinogobius sp. TO	Shounai River	River flow in pond	Aichi Expo Park, Nagakute, Aichi	18 October 2008	5	32.1–34.6
	Yahagi River	Waterway	Toyota, Aichi	8 March 2009	5	26.9–31.6
	Kushida River	Waterway	Taki, Mie	9 May 2009	5	23.1–31.7
	Yahagifuru River	Mainstream	Nishio, Aichi	18 October 2008	5	21.6–23.8
				29 April 2009	9	22.4–27.1
Rhinogobius giurinus	Yahagifuru River	Mainstream	Nishio, Aichi	29 April 2009	2	25.0, 32.2

Results and discussion

The Sr:Ca ratios measured along a transect from the center to the otolith edge of all *Rhinogobius* sp. TO samples were consistently low with mean values ranging from 0.96×10^{-3} to 2.69×10^{-3}, indicating a wholly freshwater lifestyle (Fig. 2a–d). *Rhinogobius giurinus* collected from a sympatric habitat with *Rhinogobius* sp. TO in the Yahagifuru River also had uniformly low Sr:Ca ratios throughout the otolith, with a mean value of 1.95×10^{-3} (Fig. 2e), although this fish is generally known as an amphidromous species (Tsuji 2001). This result suggests that this sampling site may be a landlocked environment in which upstream and downstream migration is obstructed by artificial steps.

In this study, none of the *Rhinogobius* sp. TO species showed sea migrating traits during the larval period, although the fish we collected may have been landlocked by artificial steps. The microhabitat environment feature of the *Rhinogobius* sp. TO inhabiting lentic water areas such as ponds, waterways, and slower current areas in river bank is fairly similar to that inhabited by *Rhinogobius* sp. OR morphotype "Shimahire" (Suzuki 1996; Suzuki and Sakamoto 2005). Tsunagawa et al. (2010) suggest that the *Rhinogobius* sp. OR morphotype "Shimahire" are freshwater residents adapting to a lentic water environment, and they do not migrate to the sea during the larval period. Furthermore, Hirashima (unpublished data) suggests that the salinity adaptation ability in *Rhinogobius* sp. TO and *Rhinogobius* sp. OR morphotype "Shimahire" is low; salinity tolerance in hatched larvae of these species are respectively 10 and 15‰, although that of *Rhinogobius* sp. OR morphotype "Shinjiko" known as an amphidromous species is 35‰. *Rhinogobius* sp. TO are also thought to be pure freshwater fish, indicating an absence of seaward migration during the larval period, like the fluvial and lentic *Rhinogobius* species.

The distributional area of the *Rhinogobius* sp. TO is limited to the waters around the Toukai region (Suzuki and Sakamoto 2005). This species is thought to have a susceptibility to human activity such as artificial river improvement and landfill of ponds, because they live in a freshwater environment their whole life, which in many cases is very close to human habitation. Therefore, conservation of river ecosystems is essential to protect the natural populations of this goby. Further studies are needed to clarify the detailed life history of *Rhinogobius* sp. TO in their entire range by ecological investigation of their distribution, reproduction, and feeding habits. Establishment



◀ **Fig. 2** Changes in the otolith Sr:Ca ratios along line transects from the core to the edge of the sagittal otoliths for each individual sample of *Rhinogobius* species. *Rhinogobius* sp. TO from **a** river flow in a pond of the Aichi Expo Park, $n = 5$; **b** a waterway from the Yahagi River, $n = 5$; **c** a waterway from the Kushida River, $n = 5$; and **d** the Yahagifuru River, $n = 14$. **e** *Rhinogobius giurinus* from the Yahagifuru River, $n = 2$

of ecological information will be useful for attempts to protect them.

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